

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

May 2014

EE403 Power Electronics & Drives - II

Date: 21.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Explain the harmonic reduction in the inverter output voltage with the help of [07]
Transformer Connections and Stepped-wave Inverters.
- Q - 2 (a) Explain in detail working of three phase bridge inverter if each of the thyristor [10]
conducts for 180° of a cycle. Also, draw phase and line voltage waveforms for a star
connected resistive load.
- (b) Explain briefly Sinusoidal-pulse modulation technique for single phase inverters. [04]

OR

- Q - 2 (a) Explain single phase Capacitor-Commutated Current Source Inverter with R load. [07]
(b) Explain working of Zero Voltage Switching Resonant Converter. [07]
- Q - 3 (a) Explain in detail Flyback switched mode power supply. [07]
(b) Explain 12 pulse converter with neat diagram and waveforms [07]

OR

- Q - 3 (a) State advantages and applications of static switches. Explain in detail single phase AC [07]
static switches.
- (b) What is the need of uninterruptible power supply? Explain Full-bridge switched mode [07]
power supply.

SECTION - II

- Q - 4 Explain with neat diagram and waveforms working of single phase half-wave and [07]
full-wave ac voltage controller.
- Q - 5 (a) Explain Integral cycle control of ac voltage controller and derive the equation for rms [07]
value of output voltage.
- (b) Explain working of single phase to single phase step up type mid-point [07]
cycloconverter and bridge-type cycloconverter.

OR

- Q - 5 (a) Explain in detail operation of three phase to single phase cycloconverter. [07]
- (b) A single phase voltage controller feeds to a resistive load of $3\ \Omega$ from 230 V, 50 Hz source. Calculate: [07]

- The maximum values of average and rms thyristor currents for any firing angle α
- The minimum circuit turn-off time for any firing angle α

- Q - 6 (a) Explain stator frequency control method for speed control of three phase induction motor. Draw speed torque characteristics of three phase induction motor with stator frequency control. [07]
- (b) Explain static Scherbius drive scheme for induction motor. [07]

OR

- Q - 6 (a) Explain working of single phase to single phase step down cycloconverter. [07]
- (b) Draw and explain speed torque characteristics of a three phase induction motor. [07]
- Explain stator voltage and frequency control of three phase induction motor.
